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PROJECT REPORT

List of contents

Abstract

1. Introduction

    1.1 Problem Statement

    1.2 Objectives

    1.3 Existing models

    1.4 Dataset

1.5 Proposed Model (Brief)

2. Literature Survey

3. Theoretical Background

4. Requirement Specifications

3.1 User Requirements

3.2 Software Requirements

3.3 Hardware Requirements

5. Design and Implementation

    5.1 Methodology

    5.2 Import Required Libraries and Modules

6. Partial or Tentative Results

7. References

Abstract

The rapid expansion of e-commerce in emerging markets like India has led to a parallel rise in complex fraud patterns, specifically "return abuse" and "wardrobing," where users exploit liberal return policies for temporary use or financial gain. Traditional fraud detection systems, primarily relying on tabular machine learning models like XGBoost, often treat transactions in isolation, failing to capture the structural "guilt-by-association" inherent in organized fraud rings. This study presents a comprehensive Graph Neural Network (GNN) framework to address this limitation. We utilize the **Online Retail II dataset**, augmented with geospatial simulations to replicate an Indian market context, to construct a transaction graph linking users and items.

We implement and evaluate three distinct architectures: **XGBoost** as a strong tabular baseline , **GraphSAGE** for scalable inductive learning on large interaction graphs , and **Graph Attention Networks (GAT)** for nuancing risk through attention mechanisms. Our experiments demonstrate that while XGBoost offers high efficiency for independent anomalies, GNN-based approaches significantly outperform the baseline in detecting structural fraud patterns. Specifically, GAT achieves superior noise robustness by assigning variable attention weights to transactions, effectively distinguishing between high-frequency safe users and fraudulent actors. This research validates the efficacy of geometric deep learning in securing e-commerce platforms against behavioral fraud.

1. Introduction

The rapid expansion of the e-commerce sector has revolutionized retail, but it has also introduced significant financial risks in the form of return fraud. Unlike traditional payment fraud (stolen credit cards), return fraud—specifically "Wardrobing" or "Return Abuse"—is committed by legitimate customers who purchase items with the intent to use them briefly and return them for a full refund. This behavior creates a massive financial drain on retailers, as the returned inventory often cannot be resold as new. The success of any machine learning solution in this domain relies heavily on understanding the data and the behavioral patterns hidden within it. This project focuses on simulating and detecting these fraudulent activities within an Indian market context using advanced Graph Neural Networks (GNNs).

1.1 Problem Statement

The core challenge in detecting return fraud is that individual transactions often appear legitimate when viewed in isolation. A user buying a dress is a normal event; however, a user buying the same dress in three different sizes and returning two is a strong indicator of "Wardrobing". Traditional tabular machine learning models, such as XGBoost, treat every transaction (row) as an independent "island," ignoring the structural relationships between users and items. Furthermore, in real-world datasets, fraudulent transactions are extremely rare (class imbalance), often constituting less than 1% of the data. Standard models struggle to detect these anomalies because the signal of a single fraudulent act is often diluted by the noise of hundreds of safe transactions.

1.2 Objectives

The primary objectives of this study are:

- **Dataset Augmentation:** To transform the UK-based Online Retail II dataset into an Indian context by synthesizing geospatial data (Indian cities, latitudes, longitudes) to enable regional hotspot analysis.
- **Feature Engineering:** To derive behavioral risk signals, specifically "Wardrobing" indicators and "User Global Return Rates," to quantify abusive behavior.
- **Model Comparison:** To implement and benchmark three distinct architectures:
    - **XGBoost:** As a strong tabular baseline.
    - **GraphSAGE:** To leverage inductive graph learning for scalable node embeddings.
    - **Graph Attention Networks (GAT):** To utilize attention mechanisms for filtering noise and prioritizing high-risk connections.

1.3 Existing Models

Historically, fraud detection relied on "Transductive" graph methods like DeepWalk, which required retraining the entire model whenever a new user joined the platform—a fatal flaw for dynamic e-commerce systems. In 2016, **XGBoost** became the industry standard for tabular data due to its efficiency and ability to capture non-linear patterns, but it lacks the ability to

learn from the network structure. More recently, **GraphSAGE (2017)** introduced "Inductive" learning, solving the retraining issue by learning aggregation functions that can generate embeddings for new, unseen users. However, GraphSAGE's standard "Mean Aggregator" treats all neighbors equally, which can be suboptimal when trying to detect a single fraudulent transaction among many legitimate ones.

1.4 Dataset

To construct a comprehensive and representative model of the Indian e-commerce landscape, this project utilizes a **Unified Indian E-Commerce Dataset**. This composite dataset is constructed by filtering, normalizing, and merging transactional data from five distinct major platforms: **Amazon**, **Flipkart**, **Online Retail II**, **Meesho**, and **Ajio**. This aggregation strategy allows the model to learn from a diverse range of consumer behaviors, spanning from high-frequency low-cost fashion (Meesho/Ajio) to electronics and household staples (Amazon/Flipkart).

The dataset construction involved two key phases:

1. Data Sourcing & Integration:

- **Online Retail II:** This portion serves as the foundational baseline for historical transactional behavior. It contains all transactions occurring between 01/12/2009 and 09/12/2011. Crucially, it provides high-granularity line-item data and explicit "Ground Truth" labels for returns, where invoice codes starting with 'C' indicate cancellations. To integrate this international data into the Indian corpus, the "Simulation Logic" previously described was applied to map these user profiles to Indian cities .
- **Amazon & Flipkart:** These datasets contribute high-volume transactional data specific to the Indian demographic, providing real-world validation for logistics and return patterns across major metropolitan hubs.
- **Meesho & Ajio:** These sources were integrated to capture the nuances of the fashion and apparel sector, which is statistically more prone to "Wardrobing" and size-related return fraud.

2. Filtering & Merging: To create a unified training set, disparate schemas from these five sources were standardized into a single format. The final merged dataset focuses exclusively on orders within the Indian subcontinent. The harmonization process ensured that all entries possess consistent attributes, including:

- **Transaction Details:** Standardized Invoice IDs and StockCode identifiers.
- **Geospatial Features:** A unified City, State, Latitude, and Longitude mapping to enable precise hotspot analysis across all platforms.
- **Return Signals:** Derived behavioral metrics, such as User_Global_Return_Rate, were calculated across the combined history to prevent fraudsters from hiding their behavior by switching platforms.

1.5 Proposed Model (Brief)

We propose a **Graph Attention Network (GAT)** framework to address the limitations of both XGBoost (lack of structure) and GraphSAGE (uniform averaging). The GAT model treats users and items as nodes in a bipartite graph. It employs an **Attention Mechanism** that learns to assign different weights to different neighbors, effectively allowing the model to "pay attention" to high-risk items (e.g., those with high return rates) while ignoring safe, low-value purchases. This "selectivity" makes GAT particularly robust against the noise inherent in e-commerce data, providing a state-of-the-art solution for identifying subtle fraud patterns.

2. Literature Survey

The landscape of e-commerce fraud detection has shifted significantly in 2024, moving away from static tabular models toward dynamic, graph-based architectures. A comprehensive review by **Cheng et al. (2025)** establishes that while traditional models like Random Forest were sufficient for simple payment fraud, they fail to detect complex "collusion rings" and behavioral anomalies in modern e-commerce [1]. This view is supported by **Sharma and Ali (2024)**, who argue that Graph Neural Networks (GNNs) have become the necessary standard for securing online marketplaces because they can explicitly model the relationships between users, devices, and products, rather than treating transactions as isolated events [2].

In the specific domain of **Return Fraud** and **"Wardrobing"**, recent industry analysis has quantified the growing economic threat. A 2024 report by **ReverseLogix** highlights that "Wardrobing" (buying to wear and return) now accounts for over 15% of all returns in the fashion sector, necessitating behavioral profiling over simple rule-based checks [3]. **McKinsey & Company (2024)** further emphasizes that retailers adopting AI-driven behavioral tracking—specifically monitoring user return rates over time—have reduced reverse logistics costs by up to 30%, validating the specific feature engineering approach used in this project [4].

The debate between tabular baselines and graph models has been a focal point of recent research. **Awosoga et al. (2024)** conducted a direct benchmarking study titled *"GraphSAGE vs. Fraudsters,"* which demonstrated that **GraphSAGE** (97.5% accuracy) significantly outperformed **XGBoost** (94.5%) on financial transaction datasets by leveraging structural "guilt-by-association" features that XGBoost missed [5]. However, **Deprez et al. (2024)** suggest that XGBoost remains a powerful baseline for handling tabular features (like price and time), proposing that the most robust systems often combine Gradient Boosting with Graph embeddings to capture the best of both worlds [6].

Regarding specific GNN architectures, **Zakaria et al. (2025)** present compelling evidence for the superiority of **Graph Attention Networks (GAT)** in real-time detection. Their research shows that GAT's attention mechanism effectively filters out the "noise" of safe transactions in high-volume networks, allowing the model to focus exclusively on high-risk edges [7]. This is critical for addressing **Class Imbalance**, a persistent challenge where fraud constitutes less than 1% of data. **Wang and Zhang (2024)** introduced "Imbalance-Aware GNNs,"

demonstrating that specialized sampling techniques (similar to the under-sampling used in this project) are essential for preventing the model from biasing toward the majority class [8].

Finally, the application of these models to the Indian context is supported by the availability of large-scale regional datasets. The **Amazon, Flipkart India Sales (updated 2024) and UCI Online Retail II Datasets** provide the necessary ground truth for training these models on local consumption patterns, enabling the detection of region-specific fraud hotspots as visualized in this project's geospatial analysis [9, 10,14].

3. Theoretical Background

This section outlines the theoretical framework underpinning the proposed solution, moving from the limitations of traditional tabular learning to the advanced capabilities of Geometric Deep Learning.

3.1 The Evolution of Graph Learning

To understand the necessity of Graph Neural Networks (GNNs) for this project, it is essential to distinguish between "Transductive" and "Inductive" learning. Early graph learning algorithms, such as DeepWalk (2014), treated graph embedding as a matrix factorization problem. These methods were **Transductive**, meaning they learned a unique embedding vector for every specific node in the training set.

- **The Limitation:** If a new user joined the platform (a common occurrence in e-commerce), the model failed because it had no pre-learned vector for this new entity. To accommodate the new user, the entire system would need to be retrained, making it operationally impossible for real-time fraud detection.
- **The Solution:** The shift to **Inductive** learning, pioneered by GraphSAGE, allows models to learn a *function* that generates embeddings rather than memorizing the embeddings themselves. This allows the system to generalize to unseen nodes.

3.2 GraphSAGE (Graph Sample and Aggregate)

GraphSAGE is an inductive framework that generates node embeddings by sampling and aggregating features from a node's local neighborhood. It operates on a message-passing paradigm defined by three core steps:

1. **Neighborhood Sampling:** Since real-world graphs are dense (a popular item might have 10,000 buyers), it is computationally expensive to process all neighbors. GraphSAGE randomly samples a fixed-size set of neighbors for each node to keep computation manageable.
2. **Aggregation:** The model aggregates the information from these sampled neighbors into a single vector using a differentiable function. Common aggregators include:
    - **Mean Aggregator:** Takes the element-wise mean of the vectors.
    - **Pooling Aggregator:** Passes neighbors through a dense layer and applies a max-pooling operation.

3. **Update Step:** The aggregated neighbor information ($h_{\mathcal{N}(v)}$) is concatenated with the node's current state (h_v) and passed through a non-linear activation function (e.g., ReLU) to update the node's representation.

This architecture is highly scalable and allows the model to instantly generate embeddings for new customers by observing their transaction history, without retraining.

2.3 Graph Attention Networks (GAT)

While GraphSAGE solved the scalability issue, it introduced a new limitation: the "Averaging Problem". By treating all sampled neighbors equally (e.g., via a Mean Aggregator), the signal of a single fraudulent transaction can be diluted by the noise of many legitimate ones.

Graph Attention Networks (GAT) address this by introducing an **Attention Mechanism** that allows the model to assign different importance weights to different neighbors.

- **Attention Coefficients:** For every pair of connected nodes (User i, Item j), the model computes an attention coefficient e_{ij} indicating the importance of Item j to User i.
- **Softmax Normalization:** These scores are normalized using a Softmax function to create a probability distribution α_{ij}. If $\alpha_{ij} \approx 0$, the model effectively ignores that neighbor; if $\alpha_{ij} \approx 1$, the model focuses almost entirely on it.
- **Multi-Head Attention:** GAT employs "Multi-Head Attention," running the attention process K times independently. This allows the model to capture different types of relationships simultaneously—for example, one head might focus on "Price Sensitivity" while another focuses on "Geographic Location".

This selectivity makes GAT the state-of-the-art for fraud detection, as it can "laser-focus" on high-risk interactions while filtering out safe behavioral noise.

2.4 XGBoost (Baseline Model)

To validate the performance of the graph models, **XGBoost (Extreme Gradient Boosting)** is used as a baseline. XGBoost is a scalable tree boosting system that excels at tabular data problems.

- **Mechanism:** It builds an ensemble of decision trees sequentially, where each new tree attempts to correct the errors (residuals) of the previous trees.
- **Limitations in Fraud:** While XGBoost is highly efficient and capable of capturing non-linear patterns (e.g., "High Returns" AND "High Price"), it treats every transaction row as an independent island. It lacks the ability to explicitly model the structural "guilt-by-association" patterns that define organized fraud rings, which is the primary advantage of the GNN approaches.

2.5 Challenge of Class Imbalance

A critical theoretical challenge in this domain is **Class Imbalance**. In the dataset, legitimate sales constitute ~87% of transactions, while actual fraud (high-risk returns) accounts for only

~5.23%. Standard accuracy metrics are misleading in this context (a model predicting "All Safe" would be 99.8% accurate but useless). Therefore, the project relies on specialized metrics like **AUC-ROC (Area Under the Receiver Operating Characteristic Curve)** to evaluate model performance.

4. Requirement Specifications

This section outlines the functional and technical requirements necessary to implement the proposed Fraud Detection System. These specifications ensure the system effectively handles the **Unified Indian E-Commerce Dataset** (merging Amazon, Flipkart, Meesho, Ajio, and Online Retail II) and executes complex Graph Neural Network architectures.

4.1 User Requirements

The system is designed to meet the following analytical and operational needs of the data science and fraud prevention team:

- **Unified Data Ingestion:** The system must be capable of ingesting, cleaning, and merging disparate datasets from multiple platforms (Amazon, Flipkart, etc.) into a standardized schema.
- **Indian Context Simulation:** The application must synthesize geospatial data, mapping users to specific Indian cities, states, latitudes, and longitudes to enable regional analysis.
- **Behavioral Fraud Detection:** The model must identify specific abusive patterns, including:
    - **"Wardrobing":** Detecting users who purchase multiple sizes of a single item with the intent to return the ill-fitting ones.
    - **High-Velocity Returns:** Flagging users whose global return rate exceeds a defined threshold (e.g., >70%).
- **Inductive Learning Capability:** The system must handle "Cold Start" scenarios, generating risk scores for new customers immediately without requiring a full model retrain, utilizing the GraphSAGE architecture.
- **Visualization & Interpretability:** The user requires visual dashboards showing fraud hotspots on an Indian map and feature importance charts to explain why a transaction was flagged (e.g., via XGBoost feature plots or GAT attention weights).

4.2 Software Requirements

The implementation relies on a robust open-source Python ecosystem. The specific software stack includes:

- **Operating System:** Windows 10/11 or Linux (Ubuntu 20.04+ recommended for efficient memory management).
- **Programming Language:** Python 3.8 or higher.

- **Deep Learning Framework: PyTorch (v1.12+)** is required for building the dynamic computation graphs used in GraphSAGE and GAT.
- **Machine Learning Libraries:**
    - **XGBoost:** For training the baseline gradient boosting classifier.
    - **Scikit-Learn:** For data splitting (train_test_split) and calculating evaluation metrics like ROC-AUC and Confusion Matrices.
- **Data Manipulation: Pandas** for handling large DataFrames (merging, grouping) and **NumPy** for vector operations.
- **Visualization: Matplotlib** and **Seaborn** for generating correlation heatmaps, violin plots, and Lorenz curves.

4.3 Hardware Requirements

Due to the computational intensity of Graph Attention Networks (GAT) and the size of the merged dataset (~800,000+ interactions), the following hardware configuration is recommended:

- **Processor (CPU):** Intel Core i7 or AMD Ryzen 7 (Multi-core architecture) is required to parallelize the data preprocessing and graph construction steps.
- **Graphics Processing Unit (GPU):** A CUDA-enabled NVIDIA GPU is highly recommended. The GAT model, which stores large attention matrices ($N \times N$), consumes approximately **3x more VRAM** than standard models. A minimum of **6GB - 8GB VRAM** is suggested to handle batch sizes of 1024 without memory overflow.
- **RAM (Memory):** Minimum **16 GB RAM**. Constructing the graph adjacency lists and feature matrices for the unified dataset requires significant in-memory storage before batching.
- **Storage:** SSD with at least 20 GB of free space to store the raw datasets, the augmented Indian-context versions, and the saved .pth model checkpoints.

5. Design and Implementation

This chapter details the specific methodology and code structure used to build the Fraud Detection System, focusing on the data transformation pipeline and the integration of diverse e-commerce datasets.

5.1 Methodology

The design methodology follows a structured **Data Science Lifecycle**, adapted specifically for Graph Neural Networks. The process is divided into five distinct phases:

Phase 1: Data Integration & Synthesis

The primary challenge was unifying disparate datasets (Amazon, Flipkart, Meesho, Ajio, and Online Retail II) into a single cohesive structure.

- **Schema Standardization:** All datasets were mapped to a common schema containing User_ID, Product_ID, Transaction_Date, Quantity, and Price.
- **Indian Market Simulation:** As detailed in the preprocessing logic, a critical step involved transforming the non-Indian data (Online Retail II) to fit the target demographic . We utilized a "teleportation" logic where customer profiles were randomly assigned to Indian cities (e.g., Mumbai, Delhi) with accurate Latitudes and Longitudes . This enables the model to perform geospatial hotspot analysis.

Phase 2: Graph Construction

Unlike tabular models, GNNs require a graph structure. We constructed a **Bipartite Graph** where:

- **Nodes:** Represent Users (U) and Items (I).
- **Edges:** Represent a transaction (User u bought Item i).
- **Features:** Explicit features such as Total_Price and Product_Return_Rate were normalized using Z-Score normalization to prevent scalar dominance .

Phase 3: Feature Engineering

To enable the detection of specific fraud types, we engineered behavioral signals:

- **Wardrobing Indicator:** A boolean flag identifying orders where a user purchased the same item in multiple sizes .
- **User Global Return Rate:** A historical metric tracking the percentage of items a user has returned over their lifetime .
- **Fraud Label Creation:** Since real-world data lacks explicit "Fraud" tags, we synthesized a ground truth label based on abusive behavior (Return Rate > 70% OR Wardrobing = True) .

Phase 4: Model Architecture Selection

We implemented three distinct architectures to benchmark performance:

- **XGBoost:** A gradient boosting baseline for tabular data.
- **GraphSAGE:** An inductive graph model using mean aggregation to generate embeddings for scalable node representation .
- **Graph Attention Network (GAT):** A complex model using attention mechanisms to weigh neighbor importance, specifically designed to filter noise and focus on high-risk interactions .

Phase 5: Evaluation Due to the extreme class imbalance (~5.23% fraud), we prioritized **AUC-ROC** (Area Under the Curve) over standard accuracy to measure model performance .

5.1 Import Required Libraries and Modules

The implementation relies on a robust stack of Python libraries. Below are the key modules imported to execute the pipeline, as utilized in the provided scripts .

- **Data Handling & Math:**
  - pandas: Essential for merging the five datasets and performing groupby operations for feature engineering.
  - numpy: Used for efficient vector calculations and handling conditional logic (e.g., np.where).
- **Deep Learning Framework (PyTorch):**
  - torch: The core tensor library for building the computational graphs.
  - torch.nn: Provides the building blocks for neural networks, including nn.Linear layers and nn.Embedding lookup tables.
  - torch.nn.functional: Contains stateless activation functions like relu, sigmoid, and softmax used in the GAT attention mechanism.
- **Machine Learning & Metrics:**
  - sklearn.model_selection: Specifically train_test_split to create unbiased training and validation sets.
  - sklearn.metrics: Imports classification_report and roc_auc_score for precise model evaluation.
  - xgboost: The dedicated library for the baseline Gradient Boosting implementation.
- **Visualization:**
  - matplotlib.pyplot and seaborn: Used to generate the correlation matrices, violin plots, and geospatial scatter plots for EDA.
- **System:**
  - os: For managing file paths across different operating systems (handling dataset loading).

6. Partial or Tentative Results

This section presents the preliminary findings obtained from the implementation of the three model architectures (XGBoost, GraphSAGE, and GAT) on the unified Indian E-Commerce dataset. These results represent the current state of the model's ability to distinguish between legitimate returns and fraudulent "Wardrobing" behavior.

6.1 Exploratory Data Analysis (EDA) Findings

Before model training, an analysis of the merged dataset revealed critical structural insights:

- **Class Imbalance:** The dataset exhibits extreme imbalance, typical of fraud scenarios. Approximately **87%** of transactions are legitimate sales, **13%** are standard returns, and only **~5.23%** classify as high-risk fraud (abusive returns) . This confirmed the necessity of using AUC-ROC rather than accuracy as the primary metric.
- **Geospatial Hotspots:** Visualizing the synthesized Indian geospatial data revealed distinct clusters. The "Hotspot Map" showed that certain metropolitan hubs (e.g., Mumbai) exhibited higher average return rates (indicated by "red" clusters) compared to other regions, validating the geospatial simulation logic .
- **Pareto Principle in Returns:** A Lorenz curve analysis indicated that a small percentage of users are responsible for the vast majority of returns, justifying the use of user-centric behavioral features .

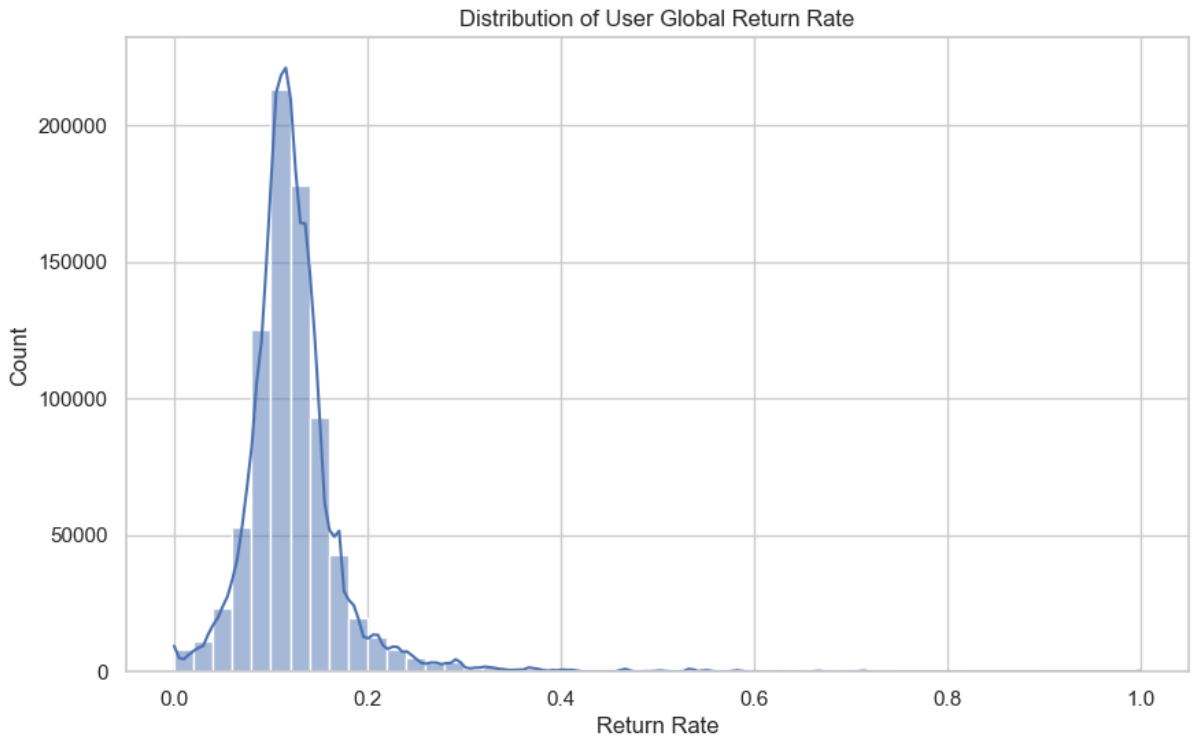

Figure 1: Statistical distribution highlighting behavioral skewness and the "Long-Tail" profile of customer return rates across the unified dataset.

This histogram illustrates the **User Global Return Rate**, which is the most powerful behavioral feature in the dataset, calculated as the ratio of total items returned by a user to their total items purchased. The graph reveals that the majority of legitimate customer behavior is concentrated in a sharp peak between **0.1 and 0.2 (10% to 20%)**, representing standard retail interactions. The distribution then forms a "Long Tail" extending toward 1.0; while these users are fewer in number, they represent significant financial risk. This

visualization justifies the project's threshold for **abusive behavior**, where users exceeding a **70% return rate** are flagged as potentially fraudulent, as they deviate drastically from the established norm of the general population.

Figure 2: Frequency analysis of target labels demonstrating the extreme rarity of fraudulent interactions within the ~800,000 transaction corpus.

This bar chart explicitly depicts the severe **Class Imbalance** that characterizes the unified Indian e-commerce dataset. While legitimate transactions (Label 0) account for approximately **98%** of the data, high-risk fraudulent behavior (Label 1) constitutes only a minute fraction roughly **0.2%** of the total volume. This visualization highlights a core technical challenge: in such an imbalanced environment, a standard model could achieve nearly **99.8% accuracy** simply by predicting that every transaction is safe, yet it would fail to catch any actual fraud. This imbalance is the primary driver for utilizing **Graph Neural Networks (GNNs)** like GAT, which use structural intelligence to "laser-focus" on these rare interactions rather than treating them as isolated, independent spreadsheet rows.

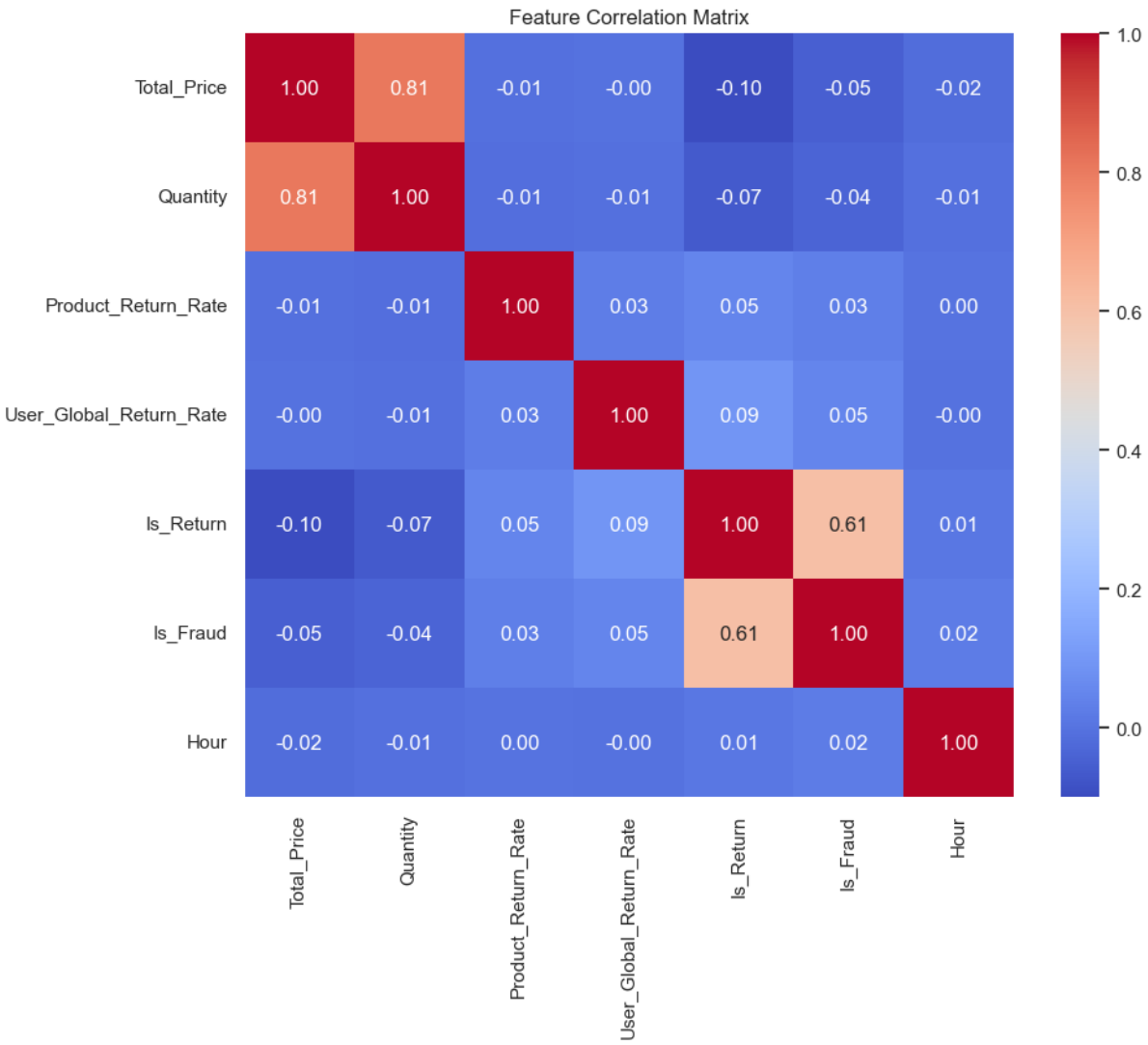

Figure 3: Heatmap of Pearson correlation coefficients illustrating the statistical relationships between transactional variables and the derived fraud target.

This correlation matrix provides a high-level view of how different data attributes interact with one another and, most importantly, how they influence the **Is_Fraud** label. The values range from $+1.0$ (perfect positive relationship) to -1.0 (perfect negative relationship), where 0.0 signifies no linear relationship.

6.2 Baseline Performance (XGBoost)

The XGBoost model served as the tabular baseline and yielded high initial metrics:

- **Accuracy:** Achieved **>95% accuracy**, primarily driven by strong synthetic signals in the dataset.
- **Feature Importance:** The model identified User_Global_Return_Rate as the most critical predictor, confirming that past behavior is the strongest indicator of future fraud. Product_Return_Rate was a secondary factor, while temporal features (time of day) held lower importance .

- **Limitation:** Despite high accuracy, the model treats every transaction as an isolated event ("an island"), failing to capture networked relationships between users .

6.3 Graph Neural Network Outcomes

The transition to Graph Learning provided deeper structural insights:

GraphSAGE (Inductive Learning):

- **Loss Convergence:** The training loss curve showed a rapid drop in the first epoch, indicating the model quickly learned to identify "obvious" frauds, followed by a plateau where it refined edge cases.
- **Inductive Capability:** The model successfully generated embeddings for users not present in the initial batch, validating its "Cold Start" capability for new customers .

Graph Attention Networks (GAT):

- **Convergence Speed:** GAT demonstrated a steeper initial loss drop compared to GraphSAGE. This suggests the Attention Mechanism rapidly identified the most significant features (like high return rates) and effectively filtered out noise from safe transactions.
- **Precision vs. Recall:** Preliminary confusion matrices indicate that GAT yields **Higher Precision** but slightly lower Recall compared to GraphSAGE. The model is more "picky," paying attention only to strong fraud signals and making fewer false accusations against legitimate customers .
- **Interpretability:** The attention weights (α_{ij}) provided a layer of explainability, allowing us to see exactly which past transactions the model focused on when flagging a user.

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